18

Lecture 18

5G Service-Based Architecture (SBA)

معماری سرویس‌محور نسل پنجم (SBA)

2G — 3G — 4G — **5G** — 6G

COURSE PART
Part III – Software-driven architectures (Lectures 15–18)

ESTIMATED STUDY TIME
6–8 hours (3 h reading, 3–5 h exercises)

DIFFICULTY
Intermediate

PREREQUISITES
Lectures 10, 14 and 16; HTTP, REST and JSON basics

In this lecture

1 Learning objectives . . . 2
2 Prerequisite knowledge . . . 2
3 Introduction and motivation . . . 3
4 Fundamental concepts . . . 3
5 Detailed technical explanation . . . 5
6 Mathematical models . . . 6
7 Network architecture and diagrams . . . 8
8 Step-by-step procedures . . . 8
9 Worked examples . . . 9
10 Technology comparison . . . 10
11 Key takeaways . . . 10
12 Exercises . . . 10
13 Further reading . . . 12

Original lecture notes for the course *Advanced Communication Networks*. The lecture sequence follows the course syllabus (26 lectures, 2G to 6G). Equations use identical notation in the English and Persian editions. Companion edition: Persian (`lecture-18-fa.pdf`).

1 Learning objectives

After studying this lecture you should be able to:

- **LO1** **Explain** the principles of the service-based architecture: NF services, producers and consumers, request/response and subscribe/notify.
- **LO2** **Describe** the protocol stack of service-based interfaces (HTTP/2, TLS, JSON, OpenAPI) and the naming and URI conventions of NF services.
- **LO3** **Trace** NF registration, discovery, authorisation and service invocation through the NRF.
- **LO4** **Compare** direct and indirect (SCP) communication and **explain** the SEPP's role in roaming.
- **LO5** **Quantify** connection needs, discovery load and latency of service-based signalling.
- **LO6** **Contrast** SBA with the point-to-point Diameter/GTP-C design of the EPC.

2 Prerequisite knowledge

You need the Diameter-based EPC interfaces (Lecture 10) as a contrast, the 5G overview of Lecture 14, the cloud-native principles of Lecture 16, and familiarity with HTTP methods (GET, POST, PUT, PATCH, DELETE), status codes, URIs and JSON.

Key terms. The vocabulary of this lecture with its Persian equivalents:

Term	Persian	Meaning in this lecture
SBA / SBI	معماری / واسط سرویس‌محور	Architecture in which NFs offer services over a common interface.
NF service	خدمت کارکرد شبکه	Capability exposed by an NF (e.g. Nsmf_PDUSession).
Producer / consumer	تولیدکننده / مصرف‌کننده	NF offering / using a service.
NRF	تابع مخزن شبکه	Registry for NF registration, discovery and access tokens.
SCP	پراکسی ارتباط خدمت	Proxy for indirect communication and delegated discovery.
SEPP	پراکسی حفاظت لبهٔ امنیت	Edge proxy securing SBI signalling between PLMNs (N32).
OpenAPI	OpenAPI	Machine-readable description of a REST API.
HTTP/2 stream	جریان HTTP/2	One request/response exchange multiplexed on a connection.
OAuth 2.0	2.0 OAuth	Authorisation framework; the NRF issues access tokens [7].
NF profile	نمایهٔ NF	Description of an NF instance stored in the NRF.

3 Introduction and motivation

The EPC was designed as a set of boxes connected by point-to-point interfaces: S6a between MME and HSS, S11 between MME and S-GW, Gx between P-GW and PCRF, each with its own protocol (Diameter, GTPv2-C) and its own peer configuration. Adding a function or a consumer meant defining a new interface and configuring every peer. In a cloud-native core, where instances come and go with load, this rigidity becomes the bottleneck.

The 5G core adopts the patterns of web-scale software: each network function exposes *services* through a common *service-based interface* (SBI) built on HTTP/2 and JSON; any authorised NF can consume them; NFs register with a repository function (NRF) and discover each other at run time [3], [6]. This lecture closes Part III by showing how SDN, NFV and web technology come together in the 5G control plane; Part IV then uses these services to build the 5G procedures.

4 Fundamental concepts

4.1 Network functions and services

The control plane of the 5G core consists of NFs connected to the SBI (Figure 1); the user plane (UPF) and the access network keep classical reference points (N2, N3, N4, N6). Table 1 lists the main NFs.

Table 1. Main 5G core network functions and example services [6].

NF	Role	Example service
AMF	Access and mobility management, NAS termination	Namf_Communication
SMF	Session management, UPF control	Nsmf_PDUSession
UPF	User-plane forwarding (not on SBI)	– (N4/PFCP)
AUSF	Authentication server	Nausf_UEAuthentication
UDM / UDR	Subscription data management / repository	Nudm_SDM, Nudm_UECM
PCF	Policy control	Npcf_SMPolicyControl
NRF	NF repository: registration, discovery, tokens	Nnrf_NFDiscovery
NSSF	Network slice selection	Nnssf_NSSelection
NEF	Exposure of network capabilities to external AFs	Nnef_EventExposure
CHF	Charging	Nchf_ConvergedCharging
NWDAF	Network data analytics	Nnwdaf_AnalyticsInfo
SCP, SEPP	Indirect communication, inter-PLMN security	– (proxies)

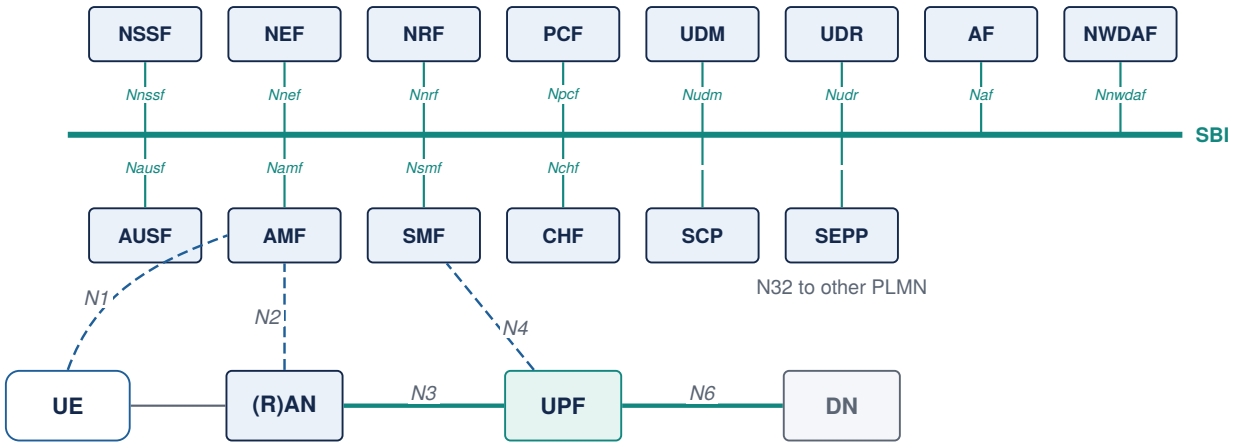


Figure 1. 5G system with service-based control plane (SBI) and reference points to the RAN and UPF.

4.2 Producers, consumers and interaction patterns

A *producer* NF offers a service; a *consumer* NF uses it. Services are named *N<nf>_<Service>_<Operation>*, e.g. Nsmf_PDUSession_CreateSMContext. Two interaction patterns exist:

- **request/response**: the consumer requests an operation and receives the result;
- **subscribe/notify**: the consumer subscribes to events (e.g. AMF subscribes to UDM for subscription changes) and the producer sends notifications to a callback URI given by the consumer.

Services are decoupled from the NFs that use them: a new consumer can use an existing service

without changing the producer – the main architectural gain over point-to-point interfaces.

4.3 Protocol stack

SBIs use HTTP/2 [9] over TCP with TLS, JSON bodies, and APIs described in OpenAPI 3.0 documents published with the specifications [2], [3]. Resources are identified by URIs of the form `{apiRoot}/<api-name>/<version>/<resource>`, e.g. `{apiRoot}/nsmf-pdusession/v1/sm-contexts`. HTTP methods map to operations (Table 2); errors are returned as HTTP status codes with a JSON problem description.

Table 2. Use of HTTP methods on service-based interfaces [2].

Method	Meaning	Typical answer
POST	Create a resource in a collection, or invoke a custom operation	201 Created (Location), 200 OK
GET	Read a resource or query a collection	200 OK
PUT	Create or replace a resource with a client-chosen URI	201 Created, 200/204
PATCH	Partially modify a resource (JSON Patch / merge patch)	200 OK, 204 No Content
DELETE	Remove a resource	204 No Content

5 Detailed technical explanation

5.1 The NRF

The NRF provides three services [1]:

- **Nnrf_NFManagement**: an NF instance registers its *NF profile* (instance ID, type, PLMN, supported slices, services and their endpoints, capacity, priority) with PUT, keeps it alive with periodic heartbeats, updates it with PATCH and deregisters with DELETE; NFs can subscribe to changes of other profiles.
- **Nnrf_NFDiscovery**: a consumer queries for producers by target NF type and criteria (slice, DNN, TAI, SUPI range, ...); the answer contains matching profiles and a validity period during which they may be cached.
- **Nnrf_AccessToken**: the NRF acts as OAuth 2.0 authorisation server and issues access tokens (client credentials grant) that a producer verifies before serving a request [7].

5.2 Direct and indirect communication

In *direct* communication the consumer sends requests straight to the producer, either from configuration (option A) or after NRF discovery (option B). In *indirect* communication (Release 16) requests pass through a *service communication proxy* (SCP): the consumer either discovers the producer itself (option C) or delegates discovery and selection to the SCP (option D) (Figure 2). The SCP plays a role similar to a Diameter routing agent or a service mesh: load balancing, routing, overload handling and observability in one place, at the cost of an extra hop.



Figure 2. Direct communication (options A/B) and indirect communication via an SCP (options C/D).

5.3 Resilience: NF sets and binding

NF instances that share context (e.g. via a common data store) can form an *NF set*; any instance of the set can serve a UE's requests, so a failed instance is replaced without losing the session. *Binding indications* tell consumers and SCPs which instance or set should receive subsequent requests or notifications for a given context. This is the SBA counterpart of the stateless-instance design of cloud-native functions (Lecture 16).

5.4 Roaming and the SEPP

Between PLMNs, NFs never talk directly: each PLMN has a *security edge protection proxy* (SEPP) at its border, and the SEPPs interconnect over N32. The SEPPs negotiate security on N32-c and protect forwarded messages on N32-f, either with TLS end to end between the SEPPs or with application-layer protection when intermediate IPX providers must be able to read or modify selected fields [5]. The SEPP thus plays the role of the Diameter edge agent of the EPC, with stronger security.

5.5 Service operations in practice

A few examples of resource-style operations (simplified) [1]:

- registration: `PUT{apiRoot}/nnrf-nfm/v1/nf-instances/{nfInstanceId}` with the NF profile;
- discovery: `GET{apiRoot}/nnrf-disc/v1/nf-instances?target-nf-type=SMF&requester-nf-type=AMF`;
- session creation: `POST{apiRoot}/nsmf-pdusession/v1/sm-contexts` with a JSON body containing SUPI, DNN, S-NSSAI and the N1 message; the SMF answers 201 with the URI of the new SM context.

6 Mathematical models

6.1 Concurrency on HTTP/2 connections

Model. A consumer sends λ requests per second to a producer; each request is outstanding for a response time R (network round trip plus processing). By Little's law the number of concurrent

requests is

$$N_{\text{inflight}} = \lambda R. \tag{1}$$

With HTTP/1.1 without pipelining, each outstanding request needs its own connection; with HTTP/2 a single connection carries up to the peer's maximum number of concurrent streams (recommended to be at least 100 [9]).

Example. $\lambda = 10^4\,\text{s}^{-1}$, $R = 5\,\text{ms}$: $N_{\text{inflight}} = 50$ – fifty HTTP/1.1 connections or one HTTP/2 connection (in practice a few, for redundancy and load spreading).

6.2 Discovery load and caching

Model. Without caching, every request triggers a discovery and the NRF sees the full request rate λ. If each of C consumers caches discovery results for K target types for a validity period T, the NRF sees approximately

$$\lambda_{\text{NRF}} \approx \frac{C\,K}{T}. \tag{2}$$

Example. $C = 100$, $K = 5$, $T = 60\,\text{s}$: 8.3 discoveries/s instead of 10^4/s (Figure 3).

Interpretation. Caching makes the NRF a light-weight directory, but cached profiles become stale: a consumer may keep sending requests to an overloaded or failed instance until the cache expires or a notification arrives. The validity period trades NRF load against reaction time.

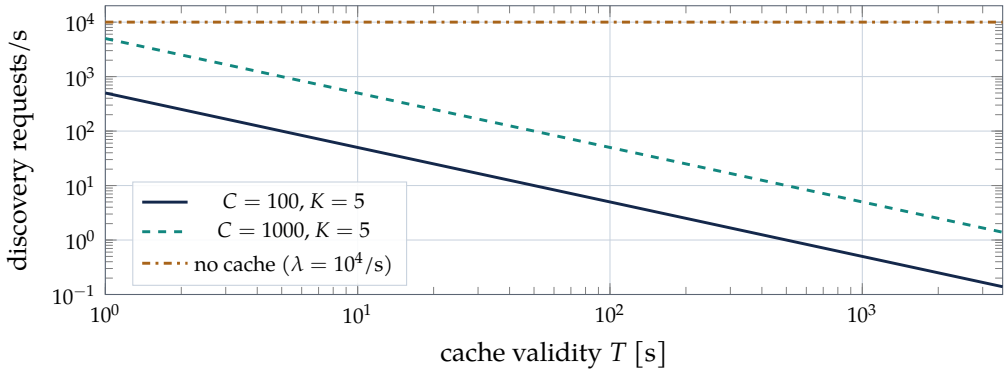


Figure 3. NRF discovery load versus cache validity (Eq. (2)).

6.3 Latency of indirect communication

Model. Let T_{dir} be the response time of a direct request. Routing through an SCP adds two network hops of one-way delay d each direction and SCP processing t_{s}; with delegated discovery the SCP occasionally (probability p_{m} of a cache miss) queries the NRF, costing T_{NRF}:

$$T_{\text{ind}} \approx T_{\text{dir}} + 2d + t_{\text{s}} + p_{\text{m}}\,T_{\text{NRF}}. \tag{3}$$

Example. $T_{\text{dir}} = 3\,\text{ms}$, $d = 0.25\,\text{ms}$, $t_{\text{s}} = 0.5\,\text{ms}$, $p_{\text{m}} = 0.05$, $T_{\text{NRF}} = 2\,\text{ms}$: $T_{\text{ind}} = 4.1\,\text{ms}$. A registration that needs ten sequential SBI exchanges gains about 11 ms.

6.4 Header compression

Model. HTTP/2 compresses headers with HPACK: the first request on a connection sends headers of H_1 bytes; later requests mostly reference table entries and cost h bytes. The mean header size

over n requests is

$$\bar{H}_n = \frac{H_1 + (n-1)\,h}{n} \xrightarrow{n\to\infty} h. \tag{4}$$

Example. $H_1 = 400\,\text{bytes}$, $h = 20\,\text{bytes}$, $n = 100$: $\bar{H} = 23.8\,\text{bytes}$ per request – a reason to keep long-lived connections between NFs.

7 Network architecture and diagrams

In a deployment, NF instances run as containers in one or more data centres; each NF type is replicated and registered in the NRF (itself redundant); SCPs or a service mesh connect them; SEPPs sit at the border; AMFs and UPFs connect to the RAN over N2 and N3. Unlike the EPC, there is no fixed peer table: who talks to whom is decided at run time from the NRF's view, the slice and the location of the UE (Figure 1).

8 Step-by-step procedures

Procedure 1 • NF registration and discovery

1. A new SMF instance starts and sends PUT with its NF profile to the NRF (Nnrf_NFManagement_NFRegister); the NRF stores it and answers 201 with the heartbeat interval.
2. The SMF sends periodic heartbeats; if they stop, the NRF marks it suspended and notifies subscribers.
3. An AMF needing an SMF for a PDU session sends a discovery GET with target type SMF, DNN and S-NSSAI.
4. The NRF returns matching profiles with a validity period; the AMF selects one (priority, capacity, locality) and caches the result.

Procedure 2 • Authorised service invocation

1. The AMF requests an access token from the NRF for scope "nsmf-pdusession" (Nnrf_AccessToken).
2. The NRF authenticates the AMF (TLS client certificate) and returns a signed token.
3. The AMF sends POST to the SMF's sm-contexts resource with the token in the Authorization header.
4. The SMF verifies the token's signature, scope and audience, creates the SM context and answers 201 with its URI; later operations use that URI.

9 Worked examples

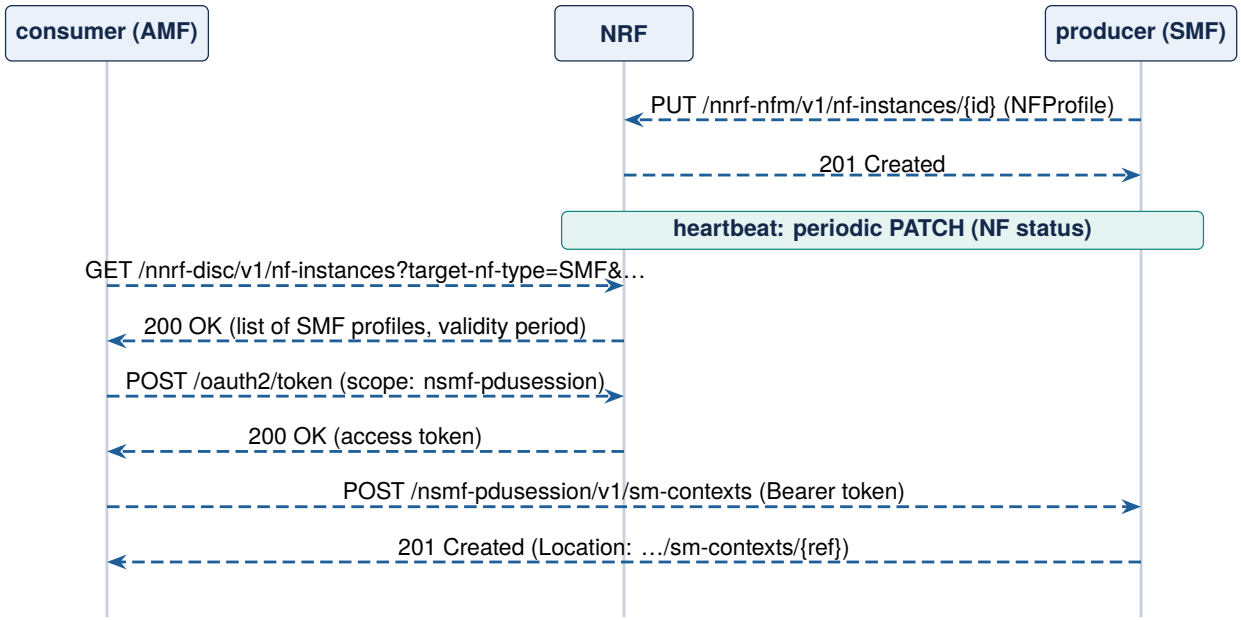


Figure 4. Registration, discovery, token and service invocation (Procedures 1 and 2).

Worked example 1 • Connections between NFs

Problem. An AMF sends 2×10^4 requests/s to a set of SMFs with mean response time 8 ms. How many requests are outstanding, and how many HTTP/2 connections are needed if each allows 100 concurrent streams and connections should be at most 50 % utilised?
Solution. $N = 2 \times 10^4 \times 0.008 = 160$. With 50 streams used per connection, at least $\lceil 160/50 \rceil = 4$ connections, spread over SMF instances for redundancy.

Worked example 2 • NRF sizing

Problem. A core has 1000 consumer instances, each discovering 5 target types with a validity of 300 s, plus 500 NF instances sending a heartbeat every 10 s. What load does the NRF see?
Solution. Discovery $1000 \times 5/300 = 16.7\,\mathrm{s}^{-1}$; heartbeats $500/10 = 50\,\mathrm{s}^{-1}$; total about 67 requests/s – small, but critical: the NRF must be highly available, because new sessions cannot be placed without it once caches expire.

Worked example 3 • SCP or direct?

Problem. With the numbers of Section 6.3, by how much does indirect communication lengthen a procedure with 6 sequential SBI exchanges? What does the operator gain?
Solution. $6 \times 1.1 = 6.6$ ms. The gains: consumers need no discovery logic or per-producer connection management; load balancing, retries, overload control and monitoring are centralised; producers can be scaled or moved transparently.

10 Technology comparison

Table 3. EPC point-to-point signalling versus 5G SBA.

	EPC (Diameter, GTPv2-C)	5GC SBA
Interface model	Point-to-point reference points	Services on a common bus; any authorised consumer
Transport	SCTP/TCP (Diameter), UDP (GTP-C)	TCP + TLS, HTTP/2
Encoding	Binary AVPs / IEs	JSON, OpenAPI-defined
Peer discovery	Static configuration, DNS, DRAs	NRF registration and discovery
Interaction	Request/answer	Request/response, subscribe/notify
Routing proxy	DRA	SCP (optional)
Roaming edge	DEA (Diameter), GTP firewalls	SEPP with N32 security
Authorisation	Implicit (peer configured)	OAuth 2.0 tokens from the NRF
Extensibility	New AVPs/IEs, new applications	New services/operations, API versions

Not everything is service-based. N1 (NAS), N2 (NGAP), N4 (PFCP) and the user plane (N3, N9 with GTP-U) keep point-to-point protocols, because they involve the RAN, the UE or high-rate forwarding where HTTP brings no benefit.

11 Key takeaways

Key takeaways

- 5GC control-plane NFs expose services (Nnf_Service_Operation) on a common SBI: HTTP/2, TLS, JSON, OpenAPI.
- Request/response and subscribe/notify; producers are decoupled from consumers.
- The NRF handles registration, discovery and OAuth 2.0 tokens; SCP enables indirect communication; SEPP secures roaming.
- Little's law sizes concurrency (λR); caching keeps NRF load at CK/T; an SCP adds about $2d + t_s$ per exchange.
- N1, N2, N4 and the user plane remain point-to-point.

12 Exercises

Exercise 1 • Level 1 – conceptual • Producers and consumers

During PDU session establishment the AMF calls Nsmf_PDUSession_CreateSMContext and the SMF calls Nudm_SDM_Get and Npcf_SMPolicyControl_Create. Identify producer and consumer in each case.

Exercise 2 • Level 1 – conceptual • Which method?

Choose the HTTP method for: (a) reading a UE's subscribed slices; (b) creating an SM context; (c) registering an NF profile with its known instance ID; (d) changing the load value in an NF profile; (e) releasing a policy association.

Exercise 3 • Level 2 – numerical • Connections

A UDM receives 5×10^4 requests/s from 40 consumer instances, response time 4 ms. Compute the concurrency and compare the number of connections needed with HTTP/1.1 (no pipelining) and HTTP/2 (100 streams per connection, one connection per consumer instance). Is one HTTP/2 connection per consumer enough?

Exercise 4 • Level 2 – numerical • Cache validity

An operator wants the NRF discovery load below $50\,\mathrm{s}^{-1}$ with 2000 consumers and 6 target types. What is the minimum validity period? What is the drawback of choosing it much larger?

Exercise 5 • Level 3 – analytical • Latency budget

A registration procedure has 12 sequential SBI exchanges with $T_{\mathrm{dir}} = 2.5$ ms. The control-plane budget for registration in the core is 50 ms. Using Eq. (3) with $d = 0.5$ ms, $t_{\mathrm{s}} = 0.8$ ms, $p_{\mathrm{m}} = 0.1$, $T_{\mathrm{NRF}} = 3$ ms, check whether indirect communication fits. What changes if NFs are in two data centres 300 km apart?

Exercise 6 • Level 3 – analytical • Design critique

Give two arguments for and two against using HTTP/2 + JSON instead of a binary protocol such as Diameter for core signalling, and explain why N2 and N4 were not made service-based.

Solutions

Solution to Exercise 1

CreateSMContext: SMF producer, AMF consumer. SDM_Get: UDM producer, SMF consumer. SMPolicyControl_Create: PCF producer, SMF consumer.

Solution to Exercise 2

(a) GET; (b) POST (to the sm-contexts collection); (c) PUT (to nf-instances/{id}); (d) PATCH; (e) DELETE (or POST to a custom "delete" operation where the API defines one).

Solution to Exercise 3

$N = 5 \times 10^4 \times 0.004 = 200$ concurrent requests. HTTP/1.1: 200 connections. HTTP/2: per consumer $200/40 = 5$ concurrent streams – one connection per consumer suffices in capacity, but two are advisable for redundancy and for spreading load over UDM instances.

Solution to Exercise 4

$CK/T \leq 50 \Rightarrow T \geq 2000 \times 6/50 = 240\,\text{s}$. A much larger validity makes consumers slow to notice new or removed producers; they rely on NRF notifications or error responses to update, and load balancing over newly scaled instances is delayed.

Solution to Exercise 5

Per exchange: $2.5 + 1.0 + 0.8 + 0.3 = 4.6\,\text{ms}$; 12 exchanges: 55.2 ms – over the 50 ms budget (direct: 30 ms). With 300 km between data centres (one-way about 1.5 ms in fibre), every cross-site exchange adds about 3 ms of round trip: the procedure must keep the NFs of one UE in one site (locality-aware discovery) or parallelise exchanges.

Solution to Exercise 6

For: widely available tools, libraries and skills; easy extensibility and versioning with OpenAPI; native fit with cloud platforms (load balancers, service meshes, observability). Against: larger messages and parsing cost than binary TLV; TCP and TLS connection management; more freedom (optional fields, versions) that can hurt interoperability. N2 and N4 involve the RAN and the user plane, which need compact, low-latency, often hardware-assisted protocols and long-lived associations; their peers are not generic consumers, so the SBA's decoupling brings little benefit there.

13 Further reading

The SBA is defined in TS 23.501 [6] and its procedures in TS 23.502 [4]; TS 29.500 [3] and TS 29.501 [2] specify the technical realisation and API design rules; TS 29.510 [1] specifies the NRF; TS 33.501 [5] covers SBA security including the SEPP. HTTP/2 is RFC 9113 [9] and OAuth 2.0 RFC 6749 [7]. Rommer et al. [8] give a textbook treatment.

References

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